



VICTORIA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
2022
HIGHER 2

NAME:

CT CLASS:

BIOLOGY

Paper 2 Structured Questions

9744 / 02

14/09/2022

2 hour

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and CT class in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in bracket [] at the end of each question or part question.

Question	Marks
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
Total	

This document consists of **26** printed pages.

[Turn over]

Answer **all** questions.

- 1 The islet of Langerhans, made up of beta cells, is an example of endocrine tissues found in the pancreas.

Fig. 1.1 shows the structure of an islet of Langerhans, and Fig. 1.2 shows the structure of a beta cell, viewed under the microscope.

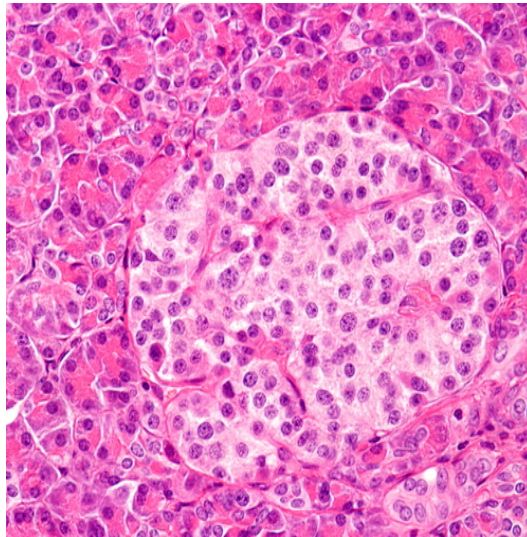


Fig. 1.1

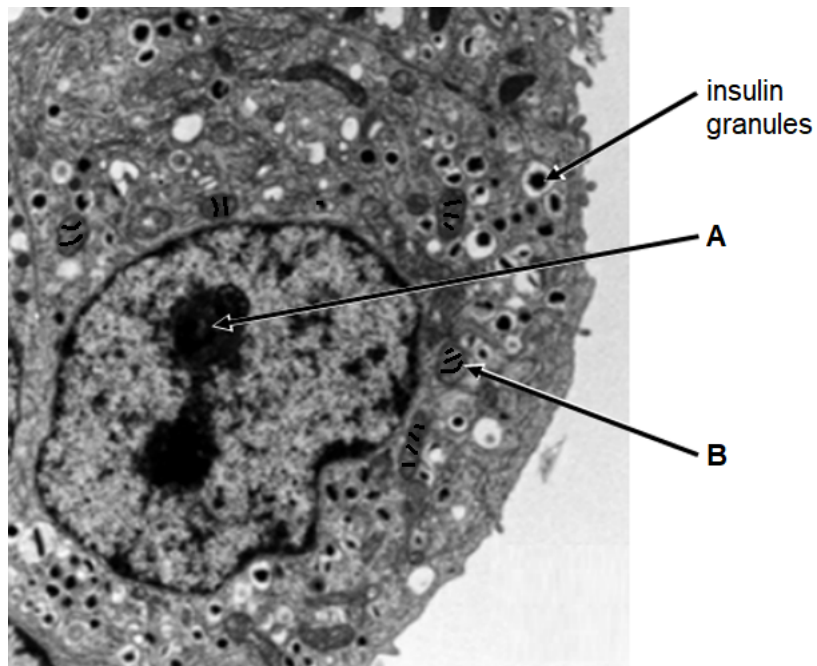


Fig. 1.2

(a) (i) Identify organelles **A** and **B** and explain their roles in beta cells.

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(ii) With reference to Fig. 1.1 and 1.2, explain how the cell theory is supported.

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(iii) Outline the roles of the intermembrane system in the synthesis of insulin within beta cells.

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(iv) Describe how insulin is released by beta cells.

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(b) Fig. 1.3 shows the structure of insulin.



Fig. 1.3

With reference to Fig. 1.3, explain how the structure of insulin is significant to its role in the body.

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[Total: 13]

- 2** Whole cow's milk contains about 87% water. The remaining 13% consists of proteins, lipids, carbohydrates, vitamins and minerals. A type of lipid found in milk is the triglyceride.

(a) Using a labelled diagram, show how a triglyceride molecule is hydrolysed.

[3]

- (b) Fatty acid β -oxidation is a multi-step process by which fatty acids are broken down to produce energy. An overview of the process is shown in Fig. 2.1.

Fatty acids primarily enter a cell via fatty acid transporters on the cell surface.

Once inside the cell, a CoA group is added to the fatty acid by fatty acyl-CoA synthase (FACS), forming long-chain fatty acyl-CoA.

The long-chain fatty acyl-CoA enters the fatty acid β -oxidation pathway, which results in the production of one acetyl-CoA from each cycle of fatty acid β -oxidation.

This acetyl-CoA then enters the mitochondrial tricarboxylic acid (TCA) cycle (also known as Krebs cycle). The NADH and FADH₂ produced by both fatty acid β -oxidation and TCA cycle are used by the electron transport chain to produce ATP.

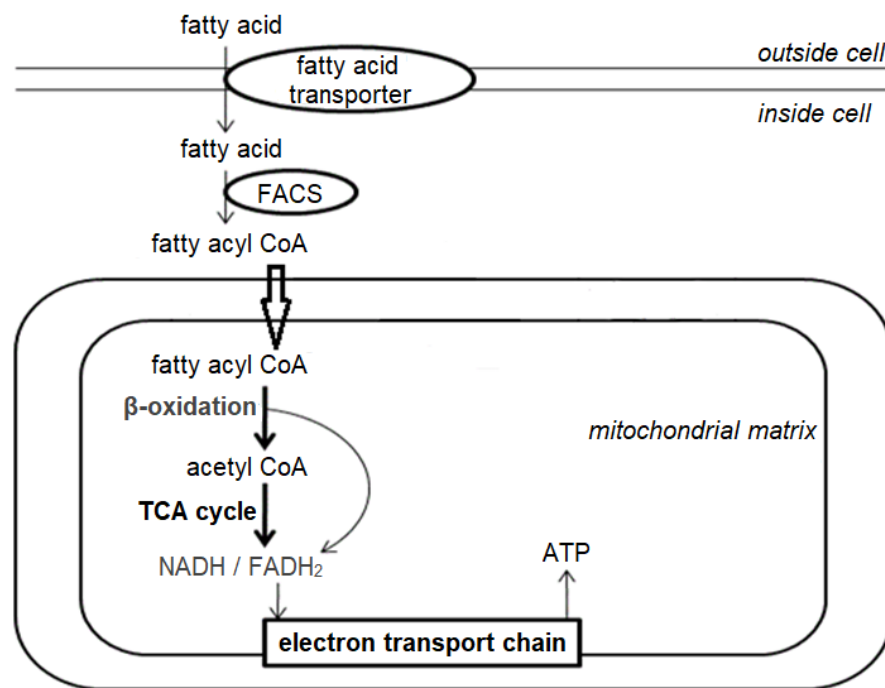


Fig. 2.1

- (i) Describe the structure of the fatty acid transporter and explain how it supports its role.

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- (ii) Account for the number of ATP molecules produced for each acetyl CoA that enters the TCA (Krebs) cycle.

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- (iii) Explain why, for the same mass of glucose and triglyceride respired, a lot more ATP is produced from the triglyceride.

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[Total: 11]

- 3 Fig. 3.1 below depicts the structure of a commonly found DNA polymerase III.



Fig. 3.1

- (a) Briefly describe the role of DNA polymerase III in a eukaryotic cell.

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- (b) Table 3.1 below shows the percentage efficiency of three different DNA polymerases at three different temperatures. The percentage efficiency is measured against the maximal efficiency of each enzyme, which is taken as 100%. The table also details the optimal pH of each enzyme.

Table 3.1

	type of polymerase	temperature			optimal pH
		30 °C	70 °C	90 °C	
percentage efficiency (%)	P-Alpha	45	93	0	7.2
	P-Beta	98	53	12	11.9
	P-Omega	26	71	96	3.2

- (i) With reference to their molecular structures, explain why the three polymerases have a different optimal pH.

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- (ii) Explain why P-Omega has a high percentage efficiency at 90 °C.

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(c) A researcher wishes to replicate DNA in a lab. He has to use one of the three polymerases indicated in Table 3.1. The buffer solution in which he has placed the DNA is highly alkaline. The procedure he wishes to employ to replicate the DNA involves heating it to 85 °C to separate the two polynucleotide strands, before lowering the temperature to 68°C to allow replication to occur.

- (i) Suggest which polymerase the researcher should choose to replicate his DNA. Explain the reasons behind your choice.

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- (ii) The three polymerases indicated in Table 3.1 are from different sources, with P-Beta being extracted from microbes living in Bonneville Salt Flats in Utah, USA.

Fill up Table 3.2 below to indicate the types of polymerase from each of the sources.

Table 3.2

type of polymerase	origin of polymerase
	yeast
P-Beta	microbes from Bonneville Salt Flats
	volcanic lake in New Zealand

[1]

[Total: 9]

- 4 The Hayflick limit, or Hayflick phenomenon, is the number of cell division a normal somatic, differentiated human cell can undergo before cell division stops.

(a) Using your knowledge on DNA replication in cells, explain how the Hayflick limit occurs.

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- (b) Differences in global levels of histone acetylation can be observed between normal and cancer cells, although the reason why cells regulate these levels has been unclear.

Fig. 4.1 illustrates the extent of histone acetylation and the corresponding intracellular pH (pHi) in the cytoplasm of the cell. Acetyl groups can be broken down into acetate and H^+ .

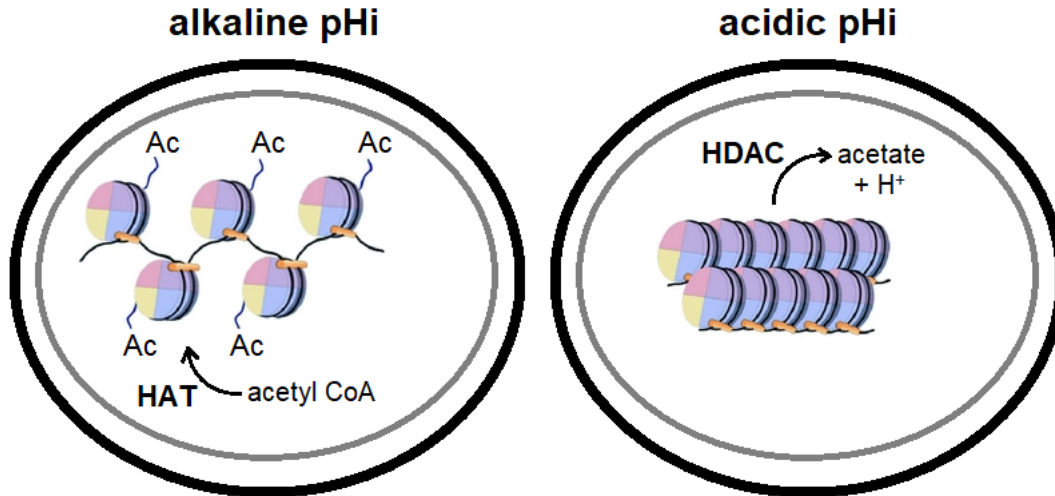


Fig. 4.1

- (i) With reference to Fig. 4.1, describe how pHi becomes acidic.

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- (ii) Suggest a role for such changes in histone acetylation other than transcriptional regulation.

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(c) Fig. 4.2 shows the synthesis and post-translational modifications of G protein-coupled receptors (GPCRs).

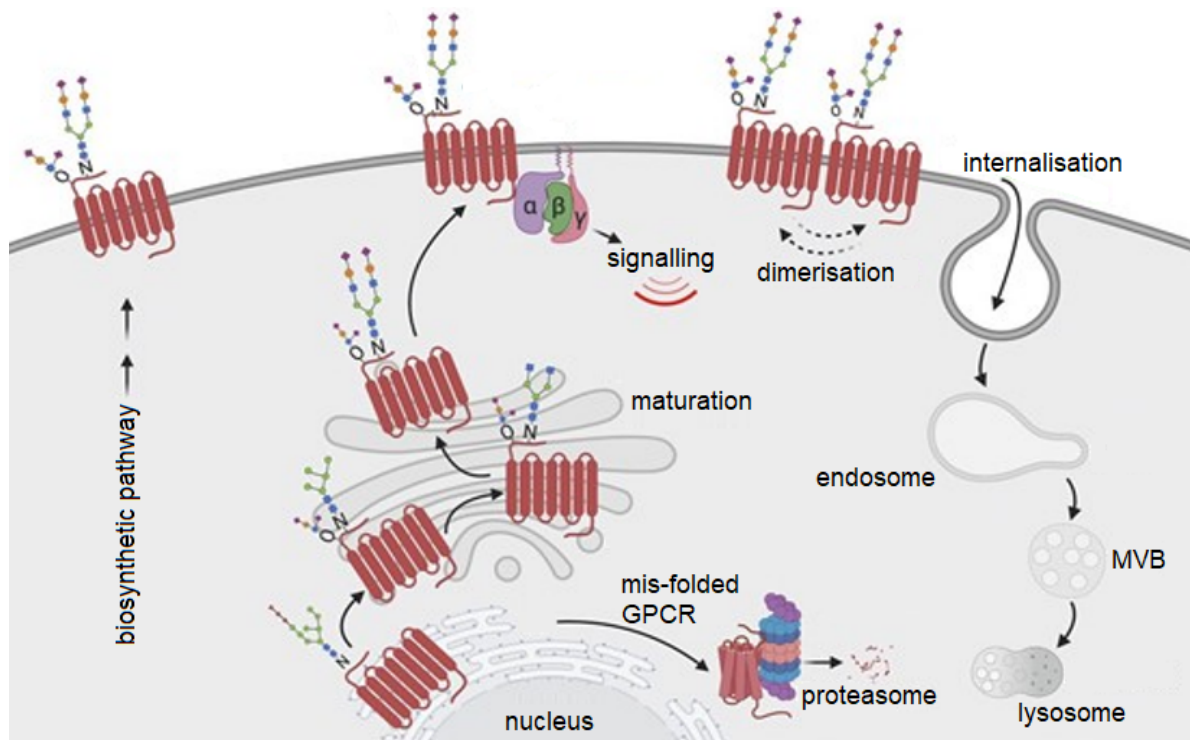


Fig. 4.2

With reference to Fig 4.2, identify and describe the post-translational control of the amount of functional GPCRs.

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[Total: 12]

- 5** Reverse transcription polymerase chain reaction (RT-PCR) is a molecular technique which combines the reverse transcription of RNA into complementary DNA (cDNA), as well as the amplification of specific DNA targets using polymerase chain reaction (PCR). It is primarily used to measure the amount of a specific RNA for analysis of gene expression.

- (a)** Account for the difference between the cDNA obtained from the RT-PCR of a eukaryotic mRNA and the gene from which the mRNA was transcribed.

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 [2]

- (b)** An experiment is conducted on several bacterial mutants to study the effect of mutations on the expression of the *trp* operon. Each mutant has a single base-pair substitution in a single component of the *trp* operon. RT-PCR is used to quantify the mRNA transcribed from the *trpE*, *trpD*, *trpC*, *trpB* and *trpA* genes of each bacterial mutant. These genes code for five enzymes E, D, C, B and A, which catalyse the synthesis of tryptophan.

The RT-PCR levels for the enzymes E, D, C, B and A in the bacterial mutants is shown in Table 5.1.

Table 5.1

enzymes	RT-PCR level for enzymes / units					
	mutant 1		mutant 2		mutant 3	
	Trp absent	Trp present	Trp absent	Trp present	Trp absent	Trp present
E	700	700	700	0	0	0
D	700	700	0	0	0	0
C	700	700	700	0	0	0
B	700	700	700	0	0	0
A	700	700	700	0	0	0

- (i) State which mutant has a phenotype consistent with a loss-of-function mutation in the *trpR* gene. Explain your answer.

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- (ii) State the component of the *trp* operon in mutant 3 in which a loss-of-function mutation has occurred. Explain your answer.

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- (iii) If the phenotype of mutant 3 is caused by a mutation in the *trpR* gene, explain how this mutation would affect the structure and function of the repressor protein.

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[Total: 10]

- 6 Studies of cell cultures of fibroblasts have revealed that progress between the phases of cell cycle is normally controlled by 'checkpoints'. If these fail, tumours may develop.

(a) The M checkpoint occurs during mitosis of the cell cycle. Explain the significance of this checkpoint.

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(b) The G₁ checkpoint functions to regulate the entry of cells into S phase, hence controlling the rate of cell proliferation. This checkpoint can be regulated by a few factors, including proto-oncogenes and tumour suppressor genes. Table 6.1 shows the results of cell counts from normal tissues of the Indian Barking Deer, in which the stages of mitosis are scored separately from cells in interphase. Table 6.2 shows the results obtained when tumour cells from the same deer are cultured.

Table 6.1

stage	number of cells	% of cells at each stage
interphase	924	87.3
prophase	46	4.4
metaphase	48	4.5
anaphase	8	0.8
telophase	32	3.0

Table 6.2

stage	number of cells	% of cells at each stage
interphase	60	42.9
prophase	17	12.1
metaphase	32	22.9
anaphase	16	11.4
telophase	15	10.7

With reference to Table 6.1, account for the differences in percentage distribution of cells at each stage.

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(c) Using information from Table 6.1 and 6.2, explain the effect of oncogenes on cell division.

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(d) Suggest why it is important to use percentage values when comparing data from the normal and tumour cell cultures.

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[Total: 7]

- 7 In a species of moth, sex is determined by the ZO system, in which there is only one sex chromosome, Z. Males have two Z chromosomes (ZZ) and females have one Z chromosome (ZO).

Two genes, **A** and **B**, are found on the Z chromosome (denoted as Z^A_B). Gene **A** codes for synthesis of a yellow pigment, giving rise to yellow body, while gene **B** codes for long antenna.

A pure-breeding male moth with yellow body and long antenna was mated with a female moth with white body and short antenna. All the F₁ moths had yellow body and long antenna. A F₁ male moth was then mated with another female moth with white body and short antenna. The following were obtained in the F₂ generation:

male moth with yellow body and long antenna	46
female moth with yellow body and long antenna	44
male moth with yellow body and short antenna	5
female moth with yellow body and short antenna	6
male moth with white body and long antenna	5
female moth with white body and long antenna	4
male moth with white body and short antenna	43
female moth with white body and short antenna	47

- (a) (i) State the genotypes of the F₁ male moth and the F₂ female moth with white body and long antenna.

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- (ii) Explain how the F₂ female moth with white body and long antenna is obtained.

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- (b) Gene **D**, located on an autosomal chromosome, codes for an enzyme that converts the yellow pigment into a brown pigment, giving rise to brown body. The recessive allele codes for a non-functional enzyme.

The F1 male moth in (a) was crossed with another female moth with white body. The following were obtained in the offspring:

male moth with brown body	48
female moth with brown body	51
male moth with white body	50
female moth with white body	51

In the space provided below, draw a genetic diagram of the above cross to show the expected phenotypic ratio of the offspring.

You do not have to include gene **B** or antenna length in the diagram.

[4]

[Total: 8]

- 8 In each species of finch, there is a variation in beak depth. Fig. 8.1 shows how beak depth is measured.



Fig. 8.1

Fig. 8.2 shows the relationship between the beak depth of parent birds and the beak depth of their offspring.

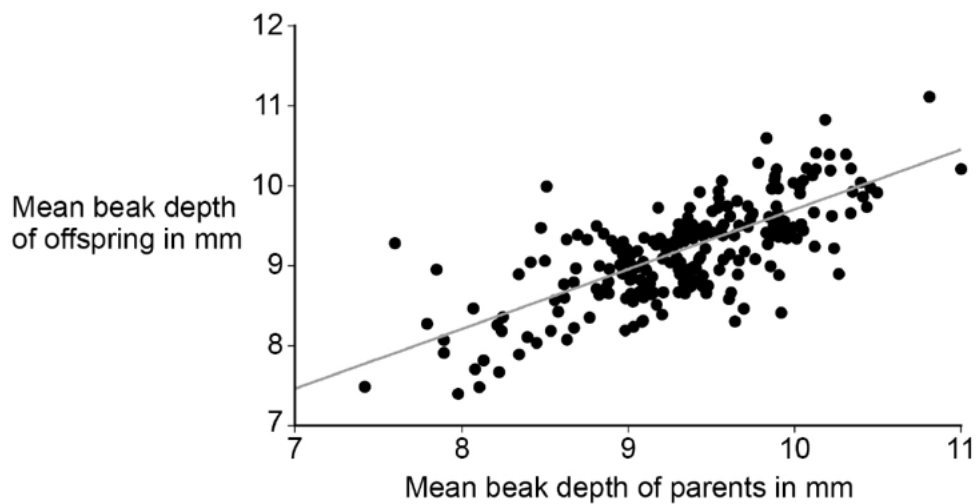


Fig. 8.2

(a) With reference to Fig. 8.2, comment whether

- (i) beak depth is an inherited characteristic.

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- (ii) more than one gene controls beak depth.

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(b) Fig. 8.3 is a map of the Galapagos Islands.

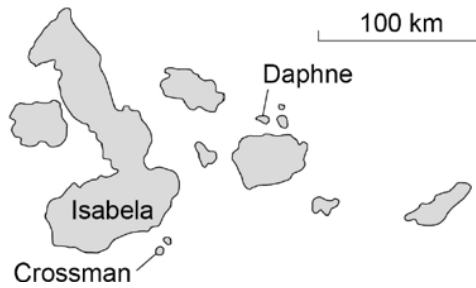


Fig. 8.3

On Isabela Island, the medium ground finch **and** the small ground finch are found. On Daphne Island, only the medium ground finch is found. On Crossman Island, only the small ground finch is found.

The medium ground finch and the small ground finch both feed on seeds. The size of seeds eaten by each bird depends on the depth of the bird's beak.

Fig. 8.4 shows how the beak depth of each species varies on each island.

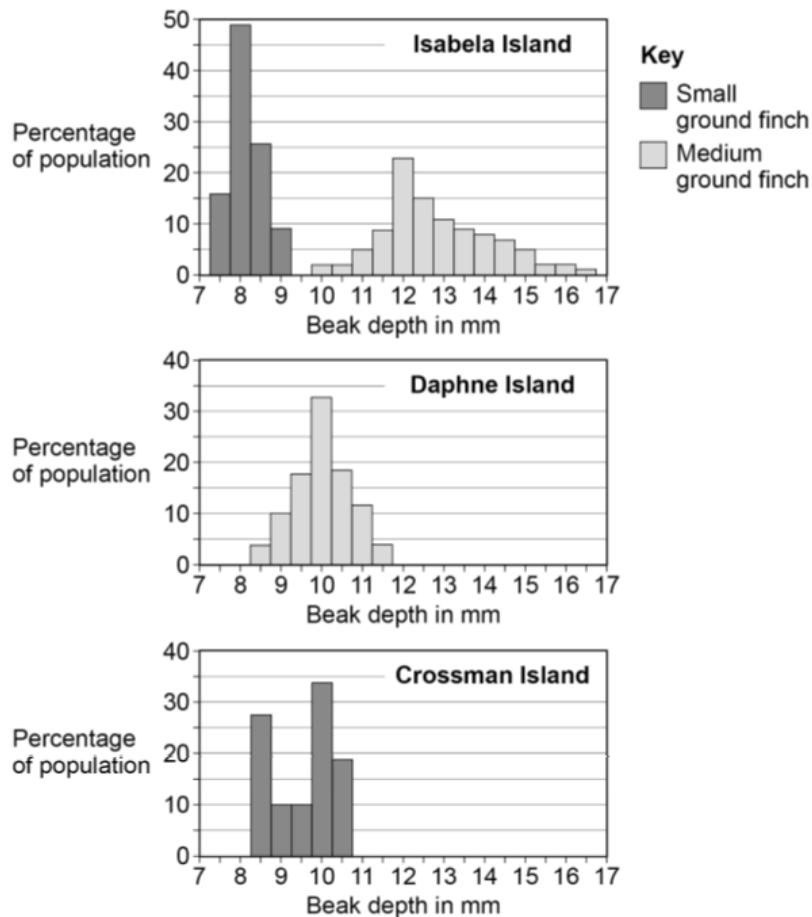


Fig. 8.4

- (i) Explain why the range of beak depth of the medium ground finch on Isabela Island is different from those on Daphne Island.

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- (ii) Outline how the medium ground finch on Daphne Island may evolve as a distinct species thousands of years in the future.

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- (iii) Suggest why both species of finches are found on Isabela Island while only one is found on the other islands.

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[Total: 12]

9 (a) Contrast between an antigen and an antibody.

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(b) Vaccination stimulates the immune system to develop immunological memory against the actual pathogen. Antigen presenting cells, such as dendritic cells, play a significant role in this process by activating specific helper T cells.

(i) Explain how the structure of the cell surface membrane of dendritic cell allows its role as an antigen presenting cell.

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(ii) Outline how immunological memory is developed after activation of specific helper T cells.

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[Total: 9]

- 10** With global warming reaching unprecedented levels, world famous iconic places like the Swiss Alps are losing the beauty of snow-capped mountain tops (Fig. 10.1).

The chart in Fig. 10.2 shows the impact of climate change on the annual temperature on the snowy mountains of Switzerland.



Fig. 10.1

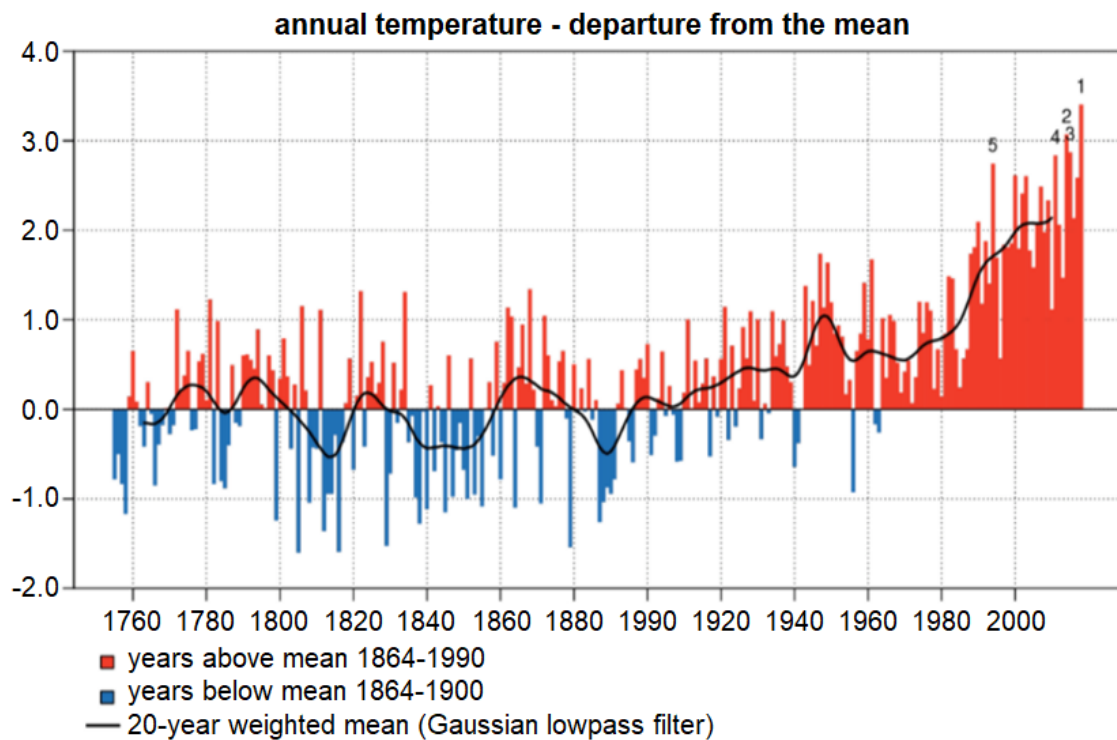


Fig. 10.2

- (a) With reference to Fig 10.1 and 10.2, other than missing ice caps, predict how the natural landscape of the Swiss Alp would change in the next decade.

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- (b) Fig. 10.3 shows the influence of increasing temperatures on the relationship between the rates of respiration and photosynthesis on stone pine seedlings found in the Swiss Alps. Gross photosynthesis refers to the carbon fixed by the plant and net photosynthesis is the measurement of carbon fixed after subtracting carbon lost via respiration (in the form of CO_2).

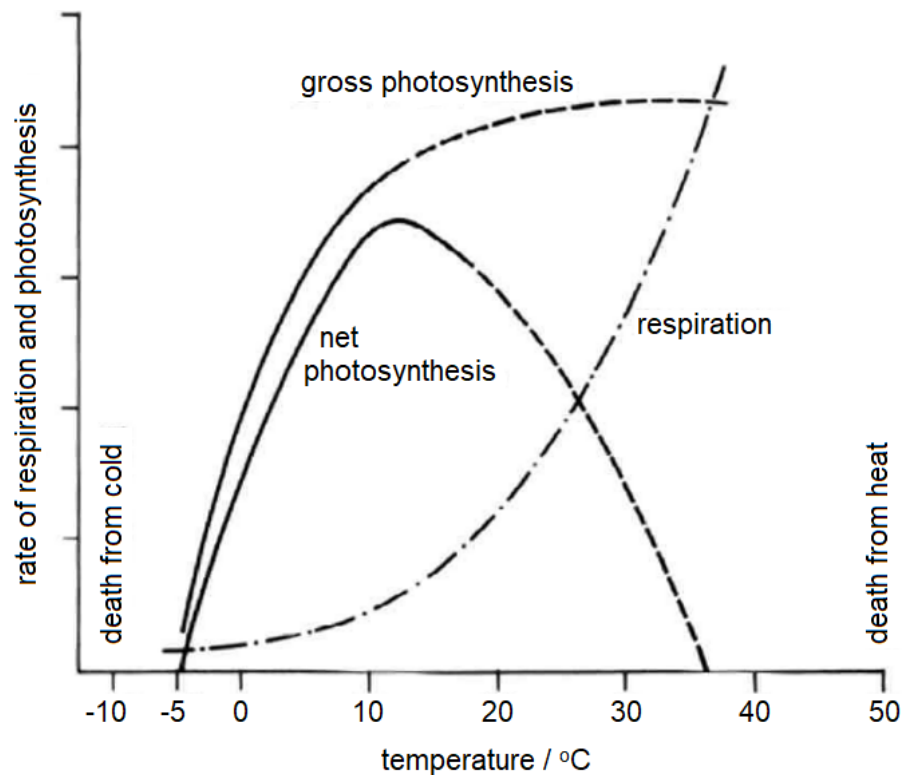


Fig. 10.3

- (i) State the optimal temperature for growth and development of the stone pine seedlings

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- (ii) With reference to Fig 10.3, describe the change in rates between the gross and net photosynthesis before 35°C.

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- (iii) With your knowledge on photosynthesis, explain the change observed in (bii).

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[Total: 9]